

direction, and five of the six paired 95% confidence intervals exclude zero. Exact sign-flip tests provide a complementary check of directional consistency. Their Holm-adjusted p -values remain above 0.05 with six independent cells. The practical interpretation therefore relies primarily on the observed effect magnitudes, uncertainty intervals, and consistent ranking direction across the available cells. Together, these statistics support a clear nominal ranking for the tested representatives while retaining the uncertainty expected from a six-cell holdout design.

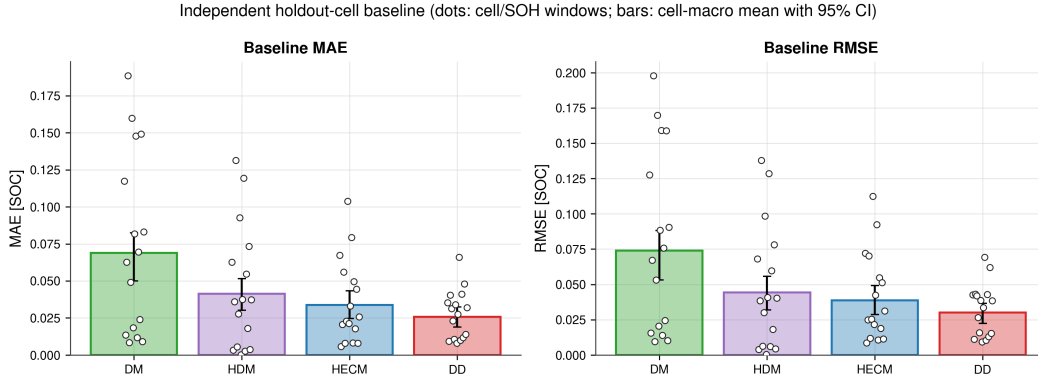


Figure 6.4: Baseline accuracy against dataset SOC across the independent hold-out test set. Dots show cell and SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

6.4.2 Current-gain sensitivity

Current-gain error is a systematic sensor error that persists over time and scales with instantaneous current. It differs from an additive offset, which shifts the measurement zero independently of current magnitude, and from zero-mean random noise. Figure 6.5 summarizes the matched signed sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the adverse-direction summary uses the larger MAE change from the two signs because one direction can partly compensate an existing baseline error.

Degradation increases monotonically with gain-error magnitude for every estimator. At 3%, cell-macro MAE rises by 0.0108 for DM, 0.0101 for HDM, 0.0060 for HECM, and 0.0038 for DD. DM and HDM integrate the scaled current directly, so the error accumulates with transferred charge. HECM is exposed through current-driven propagation, but its voltage-residual correction can counter part of the state